

University of Michigan

Interviews

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Interviews
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Executive Summary

This study explored how professionals involved in agricultural imports find, interpret, and act on regulatory information through the USDA APHIS Agricultural Commodity Import Requirements (ACIR) portal. The goal was to understand end-to-end user workflows, identify moments of confusion or friction, and learn how different roles shape the way users seek and share information.

We conducted seven semi-structured interviews lasting 45 to 60 minutes each. Participants included policy managers, PPQ officers, a training specialist, and one importer. They were recruited directly by USDA and represented multiple departments across different states. Interviews were recorded with consent, transcribed, and coded by at least two team members to identify recurring themes and contradictions.

Five key findings emerged from the data. First, the transition from PDF manuals to the current hyperlink-based format reduced scannability and made it harder for experienced users to quickly locate specific information. Second, regulatory language across commodity entries was inconsistent, creating uncertainty for users comparing requirements under time pressure. Third, contradictory classification data between different portal tiles led to incorrect regulatory decisions in the field. Fourth, the search autocomplete function caused friction by dropping results when users typed out full commodity names. Fifth, high-volume importers bypassed ACIR entirely in favor of downstream systems like e-file, suggesting the portal is not reaching one of its core user groups.

Our recommendations include offering a scan-optimized view alongside the current format, standardizing how requirement language is presented across entries, implementing cross-referencing checks to flag contradictions between tiles, improving autocomplete behavior for full-text input, and exploring tighter integration between ACIR and e-file. These findings will directly inform the tasks and focus areas for our eight upcoming usability tests, where we plan to measure task completion metrics to build on these qualitative insights.

Introduction

The Agricultural Commodity Import Requirements portal is a web-based system managed by USDA APHIS that provides regulatory guidance for importing plant products into the United States. It covers over 100,000 country-commodity combinations and serves a wide range of users, from importers and customs brokers to CBP officers and APHIS staff. Users rely on it to search for import requirements by country and commodity, access help resources, and reference documents like treatment manuals and taxonomies.

This interview study was guided by three research questions. First, what end-to-end workflows do users follow when determining import requirements for a given commodity? Second, at what points in that process do users encounter confusion or friction? Third, how do users' roles and organizational contexts shape the way they seek and share regulatory information?

Methods

Our initial stakeholder meeting helped define the project's scope and priorities. The client identified several key challenges with the ACIR portal. Users were confused by the "Pest Organism" tile and often stopped using search tools after early frustration, a pattern the client called "filter adoption failure." Some issues, like limitations with PDF search, were better addressed through training and visual cues rather than system changes. Others, like the "Woody vs. Herbaceous" seed classification conflict, pointed to a need for manual process improvements at Customs and Border Protection. The project is limited to public-facing content, which makes up 99% of portal traffic, and all recommendations must follow USDA web design standards. We cannot modify the Salesforce backend or the existing "intended use" tile structure.

Participants and Recruitment

Seven interviews were conducted with professionals involved in agricultural import processes. One interview was removed due to participant's request, and omitted from data synthesis. Five participants were USDA employees representing multiple departments across different states. One participant was a purchasing specialist at a private seed trading company that imports from roughly 40 to 50 countries. Recruitment was handled directly by USDA. While the specific selection criteria were not fully shared, all participants had experience with import regulations and the ACIR portal or related systems. Our target population included individuals across policy, field operations, training, and importing roles. We aimed for variation in job function and geographic location, though the client-led recruitment process limited our control over participant diversity.

Data Collection/Analysis

We conducted semi-structured interviews lasting approximately 45 to 60 minutes. Each participant was briefed with opening and closing scripts and informed of confidentiality protections, voluntary participation, and their right to withdraw at any time. Interviews were recorded via Microsoft Teams with participant consent. Recordings were transcribed and any identifying information was removed. Our analysis involved coding each transcript by at least two team members. Initial codes focused on recurring topics such as search strategies, interpretation challenges, and workflow inefficiencies. We intentionally looked for contradictions and differing perspectives across departments to capture variation in user experience. Findings

were generated by identifying patterns across multiple participants while noting meaningful differences.

Findings and Recommendations

Finding 1

The shift from PDFs to 508-compliant manuals reduced scannability and increased cognitive fatigue. The ACIR's transition to hyperlink-based manuals was intended to centralize information and improve accessibility. However, users described the new format as difficult to scan and navigate. One interviewee noted, "before you could just do a control F... now you have to click into a million different hyperlinks" (Interviewee #3). This change also affected training workflows. Because ACIR is now a live tool that updates continuously, trainers can no longer rely on frozen PDF copies. As one interviewee explained, "we used to train on frozen copies... but now that ACIR is a live tool, we have to update our materials a little bit more often" (Interviewee #6). We recommend creating two viewing modes for manuals. One mode would be fully 508-compliant and hyperlink-oriented. The other would be a scan-optimized, single-page view where users can globally search for terms with greater speed. This solution maintains accessibility compliance while supporting experienced users whose mental models are built around quick keyword searching.

Finding 2

Inconsistent regulatory language across commodity entries created confusion. Interviewees noted clear differences in how requirements were described. One interviewee shared that a straightforward prohibition for Rosa species from China contrasted with more ambiguous permit language for other commodities (Interviewee #2). While some entries clearly stated an item was prohibited, others required users to interpret layered or indirect language. This inconsistency increased cognitive load and created uncertainty about compliance, especially when comparing multiple commodities under time pressure. We recommend standardizing status indicators and formatting across entries to reduce ambiguity and improve decision confidence.

Finding 3

Contradictory classification data between portal tiles led to incorrect guidance. Interviewee #5 described a case where a species (*Angelonia angustifolia*) was classified differently depending on which tile was referenced. Under the plants-for-planting tile, it appeared in a category that required routing to a plant inspection station. Under the taxonomy tile, it was classified as herbaceous, meaning it did not need to be sent to the station at all. The officer initially gave the importer incorrect guidance based on one tile and had to call back to reverse the decision. The interviewee noted that "this is not the only species that's happened with" and that "certain things don't always seem the most clear" (Interviewee #5). This type of contradiction creates real

consequences for both officers and importers, leading to delays, confusion, and a loss of trust in the system's reliability. We recommend implementing a cross-referencing check that flags when classification data conflicts between tiles. An automated alert or visual indicator could prompt the ACIR team to review and resolve discrepancies before they reach end users, reducing the risk of incorrect regulatory decisions in the field.

Finding 4

Search autocomplete behavior caused friction during time-sensitive tasks. Interviewee #5 described how typing a full commodity name caused the autocomplete dropdown to disappear, forcing the user to backspace until the list reappeared. Beyond input behavior, Interviewee #2 pointed out that the portal's structure limits discovery. As a policy manager, she wanted to search broadly by country rather than always pairing a specific commodity with a country. We recommend improving autocomplete to better handle full-text input. Additionally, introducing flexible search modes that allow searching by country alone or commodity alone would better support the range of information-seeking strategies observed across user groups.

Finding 5

High-volume importers bypassed ACIR entirely in favor of downstream systems. Interviewee #4, a purchasing specialist importing from roughly 40 to 50 countries, shared that she operates "99% of the time, chiefly in the e-file portal" (Interviewee #4). She had tried ACIR several years ago but found it too complicated and stopped using it. Despite calling ACIR "the only authority for America" on import requirements, her workflow lives entirely in e-file. This suggests ACIR is not effectively reaching one of its core user groups. We recommend exploring tighter integration between ACIR and e-file so that importers can access ACIR content within their existing workflow. Importer-specific onboarding or guided search experiences could also help re-engage users who previously found the system too complex.

Discussion

Our interviews provided a nuanced understanding of the various parties who use the ACIR portal, but the study had several limitations. Our sample of six participants included primarily internal stakeholders, and we lacked perspectives from small-scale importers and first-time users. We hope to address this gap in usability testing by recruiting participants with less familiarity with the system. Additionally, the client was present during interviews, which may have influenced how openly participants responded. Their answers may have been shaped to reflect stakeholder priorities rather than personal experience. Finally, these interviews only provide qualitative context. While we identified recurring themes like filter confusion, search limitations, and regulatory ambiguity, we lacked usage analytics that would show how often and how severely these issues occur.

Our upcoming usability tests will help address these gaps. We plan to measure task completion time, error rate, and misinterpretation frequency to build a severity matrix. Since the commodity-country silo model was named as a key friction point, future research could also explore alternative search structures. These quantitative insights, combined with our interview findings, will help us prioritize redesign recommendations and directly inform the tasks and focus areas for our eight upcoming usability tests.

Appendix

Personas and Scenarios

Personas are fictional, research-based representations of key user types within a target audience. Each persona captures the goals, behaviors, frustrations, and demographic characteristics that emerged from our interview data, distilled into a single composite character. Scenarios accompany each persona by describing a realistic situation in which that person would interact with the ACIR portal, grounding the persona in a concrete task and context.

Personas and scenarios are useful for design because they shift the focus from abstract feature lists to actual human needs. They help stakeholders and designers align around a shared understanding of who will use the system and why, making it easier to evaluate design decisions against real user goals rather than assumptions.

We constructed the following three personas by analyzing themes and patterns across our six semi-structured interviews. After coding the interview data and identifying key behavioral and demographic dimensions, such as role type, frequency of ACIR use, primary task context, and level of regulatory expertise, we created characters that represent distinct segments of the ACIR user population. Each persona reflects a composite of observed behaviors and needs rather than any single interviewee. Their attributes vary along the dimensions we found most significant: whether they work in policy, field operations, or importing; how often they use the portal; and what they are trying to accomplish when they do.

Angela Rivera



ROLE

Trade Director,
International
Agricultural Policy

Background

Angela has spent over a decade working in international agricultural trade. She frequently communicates with foreign regulatory agencies and exporters who rely on ACIR to understand U.S. import requirements. She understands regulatory language deeply and often compares current ACIR entries with legacy manuals.

How She Uses ACIR

- Searches by country and commodity
- Cross-checks unclear entries against older manuals
- Clarifies ambiguous requirements for trade partners

“ The database gives me the requirement, but the manual gave me the context.

Frustrations

- ACIR feels too siloed by commodity and country
- Cannot search by general regulatory concept (e.g., definitions across commodities)
- Feels some nuance was lost when manuals were consolidated

Primary Goal

Ensure agricultural trade between countries complies with U.S. regulations while maintaining clarity and consistency across stakeholders.

Core Needs

- Concept-based search
- Linked regulatory definitions
- Better traceability from legacy manual language

Marcus Bennett



ROLE

Operations Manager,
Agricultural Import
Company

“ The database gives me the requirement, but the manual gave me the context.

Primary Goal

Determine quickly whether a product can be imported and what documentation is required.

Background

Marcus manages logistics and supplier coordination for a mid-sized seed and plant distributor. He's business-savvy but not trained in regulatory interpretation. He previously struggled with dense PDF manuals and avoids complex filtering systems because they feel overly technical.

How She Uses ACIR

- Searches specific commodities
- Looks for simple admissibility confirmation
- Skips advanced filters

Frustrations

- Regulatory language is dense
- Filtering system feels confusing
- Too many clicks to reach an answer

Core Needs

- Plain-language summaries
- Guided decision pathway
- Clear checklist of required steps

Claire Thompson



ROLE
PPQ Officer, Field Operations (Plant Inspection Station)

Background

Claire works on the front lines reviewing imports daily. She makes regulatory decisions that directly impact shipments at ports of entry. She is highly familiar with ACIR and uses it regularly, often under time pressure.

How She Uses ACIR

- Searches commodity and country combinations
- Cross-checks taxonomy and admissibility tiles
- Interprets seed limits and treatment requirements

Frustrations

- Inconsistencies between tiles (e.g., taxonomy vs. admissibility)
- Ambiguity in wording (e.g., "small lot of seeds" limits)
- Must explain nuanced regulatory language to non-experts

Primary Goal

Quickly determine admissibility requirements and provide accurate guidance to brokers and importers.

Core Needs

- Conflict alerts between tiles
- Embedded clarifications and definitions
- Real-world examples within entries

Quote

"I just need it to clearly say yes or no, and why."

Angela Rivera: Angela searches ACIR to interpret and validate an official commodity-country import requirement before advising an international trade partner on compliance expectations.

Marcus Bennett: Marcus uses ACIR to determine whether a shipment at the port is admissible, requires inspection at a plant inspection station, or must be held due to specific regulatory conditions.

Claire Thompson: Claire uses ACIR to cross-check taxonomy and admissibility tiles under time pressure at the port, resolving conflicting classification information so she can give brokers and importers a clear yes-or-no answer on whether a shipment can be released.

Interview Protocol

Introduction

Hi, my name is [Name], and I'm part of a student team from the University of Michigan School of Information. We're conducting this interview as a part of our course project to explore how professionals involved in agricultural imports find, interpret, and use regulatory information in their day-to-day work. We're especially interested in understanding your experiences navigating online systems that provide this type of information.

This interview will last approximately 45 minutes to an hour. During that time, I'll ask a series of questions to better understand your perspective on these topics. Please feel free to respond based on your own experiences and insights. I'm here to learn from you.

Before we begin, I want to cover a few important points about your participation. This session will be audio and/or video recorded for analysis purposes only. Your name will not appear in any report, and recordings will be deleted after the project ends. We may include anonymous quotes in our findings, but nothing will be traceable back to you. If at any point there is something you prefer not to be on the record, even anonymously, please let me know.

Your participation is completely voluntary. You may skip any question, stop the interview at any time, or request that any information you've shared be removed, with no consequences. If you have any questions about this study, you can contact our research team via email included in the schedule.

Before we get started, do you have any questions? And can I confirm that you've signed the consent form?

Questions

1. Tell me about yourself and your background in the industry.
2. Tell me about your roles and responsibilities related to importing agricultural commodities.
3. What types of decisions are you responsible for when preparing or reviewing import requirements?
 - a. If anyone, who else is typically involved in this?
4. Can you walk me through the last time you searched for something on the ACIR portal? What did you search for, and how did you narrow down the results?

- a. When you've been unsure about something on the portal, what did you do? Did you look for help content, ask a colleague, or try something else?
5. What does a “successful” outcome look like for when you’re gathering information?
6. What kind of information or data do you rely on most when preparing for an import?
 - a. Where do you typically find this information?
 - b. How do you determine if the information is up to date?
7. What challenges do you encounter in your daily work and process?
 - a. Are there points that feel repetitive?
 - b. Have you ever had to redo work due to incorrect guidance or unclear instructions?
 - c. Are there points where you feel uncertain of your understanding
8. Have you ever needed to look up import requirements while away from your desk or in the field? What was that experience like?
 - a. In what settings do you access the information?
9. How important is speed versus comprehensiveness of this platform for you?
10. Do you use any systems alongside the ACIR portal?
11. How do you document or report your findings to others?
12. If you could redesign or improve the process of finding import requirements, what would you do?

Conclusion

Thank you so much for taking the time to share your insights, experiences, and perspectives with me today. Those are all the questions I have for now, but I’d like to leave the floor open in case there’s anything else you’d like to add that we didn’t cover.

As a next step, your feedback, along with input from other participants, will be reviewed and synthesized to identify common themes and patterns. These insights will help inform the direction of our project and potential solutions. I may reach out with a few follow-up questions as we move forward; would email be the best way to contact you?

As mentioned at the beginning of the interview, your identity will remain confidential when your feedback is shared with the team. Before we wrap up, I want to confirm that you’re comfortable with all the information you’ve provided being included. Is there anything you’d like to exclude or clarify?

Thank you again for your time and for sharing your valuable perspectives, I truly appreciate it.

Consent Form

Interview Consent Form

Study: UX Research on Online Agricultural Import Information Systems

Researchers: University of Michigan School of Information Graduate Students in SI 622

Shenghao Mei Seorin Jang Dongsub Kim Micaela Ciambrone Layilah Campbell

Purpose

We are a team of graduate students in the University of Michigan's School of Information working on a project in partnership with the Information Services and Manuals Unit of APHIS to explore the usability of the Agricultural Commodity Import Requirements (ACIR) database.

What to Expect

This interview will take about 45-60 minutes. We will ask questions about your experiences finding regulatory or import-related information online and how it intersects with the Agricultural Commodity Import Requirements database. The session will be video recorded and transcribed.

Your Rights

Your participation is voluntary. You can skip any question or stop at any time without penalty. You will not be paid for participating.

Confidentiality

Your name will not appear in any report. Recordings will be used only for analysis by our research team and will be deleted after the project ends. We may include anonymous quotes in our findings. Don't share any information meant for official use or which is proprietary trade information.

Questions

If you have questions, contact the research team at: mciambro@umich.edu or the UMSI team edward.metz@usda.gov

By signing below, I confirm that:

- I have read and understood this form
- I agree to participate in this interview

- I agree to be recorded

_____	_____
Participant Signature	Date

Printed Name